DHCP and Cisco Packet Tracer Assignment

Complete the practical tasks in Cisco Packet Tracer and attach screenshots where indicated.

1. DHCP Server with Two Client PCs

Create a network with 1 DHCP Server, 1 Switch, and 2 PCs. Configure a DHCP pool and verify that both PCs receive different IP addresses automatically.

Device	Assigned IP Address
PC-1	_____
PC-2	_____

2. DHCP DORA Sequence Diagram

```
Client PC                    DHCP Server
    |---- DHCP Discover ----->|
    |<----- DHCP Offer -------|
    |---- DHCP Request ------>|
    |<--- DHCP Acknowledge ---|
```

Label the sender and receiver for each message in your final diagram.

3. DHCP Pool with Only One Available Address

Change the DHCP pool so it contains only one available IP address. Connect two PCs and observe the result.

Screenshot Placeholder: _____

Explanation: The DHCP server has only one usable IP address in its pool. The first PC receives the address, while the second PC fails to obtain an IP because no addresses remain available for allocation.

4. Zomato-Style Restaurant WiFi Scenario

Configure DHCP for a Laptop and Smartphone (represented by a generic PC) connected to a restaurant WiFi network.

Device	Assigned IP
Laptop	_____
Smartphone / Generic PC	_____

Why DHCP is Useful:

DHCP automatically assigns IP addresses and network settings to devices joining the WiFi network. This reduces manual configuration and helps multiple customers connect quickly and reliably.

Submission Checklist

- Screenshot of DHCP configuration
- Screenshot showing both PCs receiving IP addresses
- Screenshot showing DHCP pool exhaustion
- Screenshot of restaurant WiFi scenario
- Short explanation of observations